



Constructed wetlands for the removal of pharmaceuticals and personal care products (PPCPs) from wastewater: origin, impacts, treatment methods, and SWOT analysis

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Abstract The continuous exposure to pharmaceuticals and personal care products can lead to a series of individual antagonistic and synergistic effects and long-lasting toxicity to humans and aquatic lives. This may also lead to developing antibiotic resistance, teratogenic, carcinogenic, and endocrine-disrupting effects. However, several PPCPs are also considered biologically active for non-target aquatic organisms, such as mosquito fish, goldfish, and the algae

Pseudokirchneriella subcapitata. Various physico-chemical methods such as ozonation, photolysis, and membrane separation are recognized for the effective removal of PPCPs. However, the high operation and maintenance costs and associated ecological impacts have limited their further use. Constructed wetlands are considered eco-friendly and sustainable for the removal of pharmaceuticals and personal care products together with antibiotic resistance genes. Several mechanisms such as sorption, biodegradation, oxidation, photodegradation, volatilization, and hydrolysis are occurring during the phytoremediation of PPCPs. During these processes, more than 50% of PPCPs can be eliminated through constructed wetlands. They also offer several additional benefits as obtained macrophytic biomass may be used as raw material in pulp and paper industries and a source for second-generation biofuel production. In this study, we have discussed the origin and impacts of PPCPs together with their treatment methods. We have also investigated the strengths, weaknesses, opportunities, and threats associated with constructed wetlands during the treatment of wastewater laden with pharmaceutical and personal care products.

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Introduction

An increasing number of waterbodies are gradually losing their ecological values due to the deterioration of their water quality through several anthropogenic and industrial activities (Bi et al., 2019). Around 54% of the world's population lived in urban areas as per the estimates of 2014, and it is predicted to increase up to 66% by the year 2050 (Jensen & Wu, 2018). The enormous growth in population leads to high medication demand for humans, disinfectants, veterinary drugs, fragrances, and other domestic compounds (Rajapaksha et al., 2019). These pharmaceutical and personal care products (PPCPs) frequently comprise a huge number of chemicals with dissimilar physio-chemical characteristics (Ebele et al., 2017). They contain pharmaceutical products such as antibiotics, painkillers, anti-inflammatory and anti-allergic drugs, β -blockers, and hormones that are utilized to treat illnesses in both human beings and animals (Chopra & Kumar, 2018). However, several personal care products such as deodorants, makeup, sunscreens, moisturizers, hand soaps, and insect repellants are used mainly to improve the well-being and living standards (Abdullah et al., 2020). The majority of the PPCPs are discharged into the environment via improper disposal and release from manufacturing locations which are ultimately transported to fresh waterbodies and disturb the whole ecosystem (Li et al., 2019). The maximum number of pharmaceutical chemicals are explicitly considered to be soluble in water adequately and habitually supposed to react alike. They are usually passed over the digestion system of humans as well as animals before reaching the environment. Continuous exposure to PPCPs can lead to individual antagonistic and synergistic effects and also long-lasting toxicity such as antibiotic resistance and teratogenic, carcinogenic, and endocrine-disrupting effects on humans and aquatic life (Sutar et al., 2019). Therefore, the contamination caused by PPCPs has been considered a serious hazard to the aquatic ecosystems; nevertheless, unfortunately, no regulatory standards for the effluent released from manufacturing units and wastewater treatment plants (WWTPs) have been established to date (Shetty & Gupta, 2018). Most of the PPCPs find their way to freshwater through WWTPs, particularly from healthcare facilities (Hu et al., 2021; Mompelat et al., 2009). The WWTPs can eliminate

PPCPs; though their performance varied critically depending upon their physicochemical properties as well as ecological settings (Li et al., 2016). Overall, the effectiveness of WWTPs for the exclusion of PPCP is considered insufficient (Ramírez-Morales et al., 2020; Rout et al., 2021). Several conventional technologies such as adsorption, membrane filtration, and advanced oxidation have been identified with high potential for the removal of residual PPCPs. However, the high energy and chemical requirements and associated ecological impacts restrict their use more frequently (Li et al., 2019). Consequently, despite the technologies existing for the removal of PPCPs, there is still an increasing trend to develop a low-cost, eco-friendly, and efficient alternative for the treatment of wastewater containing a diverse range of PPCPs (Kaur et al., 2019). Constructed wetlands (CWs) are considered sustainable and environmentally friendly options as they need lower-cost investment and greater removal effectiveness for different PPCPs (Xu et al., 2015; Sgroi et al., 2018; Zhang et al., 2018a; Al-Baldawi et al., 2021; Hu et al., 2021). They essentially depend on the natural ongoing processes in which macrophytes and associated microbial assemblages effectively degrade/transform various PPCPs (Kumar et al., 2020, 2021; Tripathi et al., 2020). Additionally, metabolic degradation or transformation may achieve complete mineralization and produce CO_2 and H_2O . CWs provide a favorable atmosphere to sustain the growth of microbes such as yeast, bacteria, algae, and fungi that helps in the biodegradation process (Fadhil & Al Baldawi, 2020; Meng et al., 2021; Wan et al., 2021). CWs are studied to have great potential for the elimination of several pollutants through various physicochemical and biological methods (Gorito et al., 2017; Wan et al., 2021). They can achieve advanced removal efficacies under optimized conditions, especially for vegetation, substrate materials, microbial communities, hydraulics, etc. (Dordio & Carvalho, 2018; Kumar et al., 2020; Wu et al., 2015). Till now, several review articles have been published on the elimination of PPCPs through CWs, but they are mainly concerned with specific features of the wetland process (design, origin, fate treatment, and impact) (Chen et al., 2018; Al-Baldawi et al., 2021; Hu et al., 2021). However, the articles dealing with a comprehensive review of several aspects of the removal of PPCPs are not more frequent. This review

may offer additional information about the source and impacts of PPCPs on humans as well as aquatic environments throughout the globe. This study also strengthens the existing treatment methods with a strong focus on CWs as an efficient and alternative technology for the elimination of PPCPs. Lastly, we executed the strengths, weaknesses, opportunities, and threats (SWOT) investigation of CWs to evaluate their effectiveness and sustainability for the exclusion of PPCPs to safeguard the environment.

Occurrence and potential source of PPCPs in the environment

A huge amount of different PPCPs is generated throughout the globe and discharged into the environment as a by-product of regular lifestyle and well-beings (Fig. 1) (Noguera-Oviedo & Aga, 2016). However, the prime sources of PPCPs included individual household activities, WWTPs, manufacturing units,

landfill sites, large agricultural farms, etc. (Chopra & Kumar, 2018). During the therapeutic period, several pharmaceutical compounds that are taken up later are excreted into sewage systems and finally reach the aquatic ecosystem (Dey et al., 2019). However, the concentration of specific PPCPs in municipal wastewater is strictly dependent upon releases from nearby manufacturing units, total inhabitants, usage patterns, and inappropriate dumping of unused or expired drugs. Several interconnected social and cultural factors such as improper usage, the weak financial status of families, non-enforcement of regulating laws, low education and expertise, and extensive husbandry management schemes impact the usage patterns of PPCPs, and the amount varies depending upon time and region (Eggen & Vogelsang, 2015). It is reported that effluents generated from the hospitals possess higher per capita loads of several antibiotics, anti-inflammatories, analgesics, and antitumor drugs in addition to pharmaceuticals associated with the laboratory and diagnostic activity (Alhaji

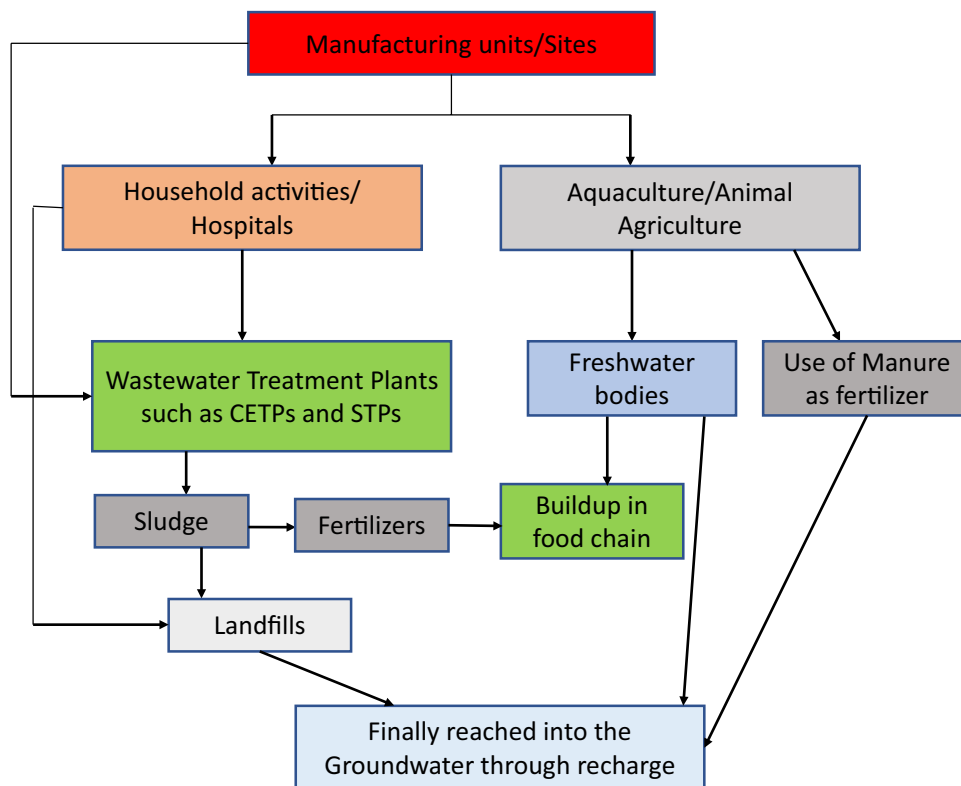


Fig. 1 Potential sources and routes of PPCPs within the aquatic ecosystem (modified from Al-Baldawi et al., 2021)

& Isola, 2018). The raw sewage of several European cities has revealed distinct spatial and temporal patterns in illicit drug usage (Ort et al., 2014; Thomas et al., 2012). A higher concentration of cocaine and ecstasy was detected on off days as compared to working days. The concentration of antibiotic chloramphenicol utilized to treat respiratory tract infections was recorded higher in winter as compared to summer, whereas the concentration of some insect repellents and sunscreen agents such as biocides, N, N-diethyl toluamide, benzophenone-3, and octocrylene has been found higher during the summer seasons (Eggen & Vogelsang, 2015). It is reported that the complex arrangement of PPCPs restricts their removal completely through conventional treatment facilities, and hence the availability of several PPCPs in treated water has been detected worldwide (Chopra & Kumar, 2018). However, the generation of various metabolites through transformation is also recognized during the removal process through WWTPs (Tarpani & Azapagic, 2018). Disposal of expired drugs from personal uses together with domestic waste is another probable way to contaminate nearby waterbodies (Hoyett, 2018). The bio-transformed PPCPs may percolate underground and pose serious threats to groundwater (Ebele et al., 2017). Animal agriculture and aquaculture are other important sources of several therapeutic antibiotics and artificial hormones that act as regulators for growth and reproduction in animals and aquaculture (goat, chicken, cow, crabs, fish, shrimps) (Noguera-Oviedo & Aga, 2016). These contaminants may finally end in the surface waters or might reach in the soil by animal feces and percolate into aquifers (Dey et al., 2019). It is evaluated in the previous studies that PPCPs are also discharged into sewage networks via regular household washing activities (Rodriguez-Narvaez et al., 2017; Yang et al., 2017). However, the dissimilar distribution patterns of antibiotics such as quinolones, azithromycin, and sulfonamides in sediment and water was studied by Ngo et al. (2021). They also explained that the compounds such as sulfamethoxazole with a low octanol–water partition coefficient (Kow) have fewer sediment sorption capacities. In addition, azithromycin with high Kow reveals excellent sorption capacity on sediments. Furthermore, low volatility, high polarity, and hydrophilic characteristic limit their dispersion in the environment (Ebele et al., 2017). Traditional WWT systems such as

stabilization ponds are exhibited as a potential source of PPCPs in water streams as prescribed by K'oreje et al. (2018). The occurrence of PPCPs in fresh waterbodies has been documented regularly throughout the world. E.g., in Shanghai, China, several cardiovascular drugs and antibiotics were recorded up to 15% and 58%, respectively, in tap water (Liu et al., 2019a) in which phenicols (94%) were the most dominant. The concentrations of quinolones and bisphenol A (estrogens) have been also recorded higher among other PPCPs in several waterbodies in China (Xu et al., 2019a). However, Li et al. (2018) reported that along the Songhua River, caffeine and triclosan were the most dominant drugs, signifying the diverse fates of PPCPs. Presence of lipophilic medicines such as diclofenac, erythromycin, propranolol, indomethacin, sertraline, ketoconazole, carbamazepine, gemfibrozil, and 17 α -Ethinyl estradiol bezafibrate in fish tissues, sediment, suspended matters, colloidal phase, and dissolved phase water demonstrates their higher bio-availability (Liu et al., 2018a). In the Ganges River of India, the concentrations of caffeine and ketoprofen were recorded as the maximum among others (Sharma et al., 2019). Similarly, in Brazil, caffeine was predominant in both Anil and Bacanga Rivers as described by Chaves et al. (2020). However, the concentrations of diclofenac, ibuprofen, carbamazepine, sulfamethoxazole, and acetaminophen were also recorded higher. Archer et al. (2017) recorded the presence of three illegal drugs, namely, cocaine, mephedrone, and methamphetamine in the surface water of South Africa. However, the high concentrations of PPCPs such as methylparaben, lamivudine, sulfamethoxazole, and zidovudine were also detected in rivers of Kenya (K'oreje et al., 2018).

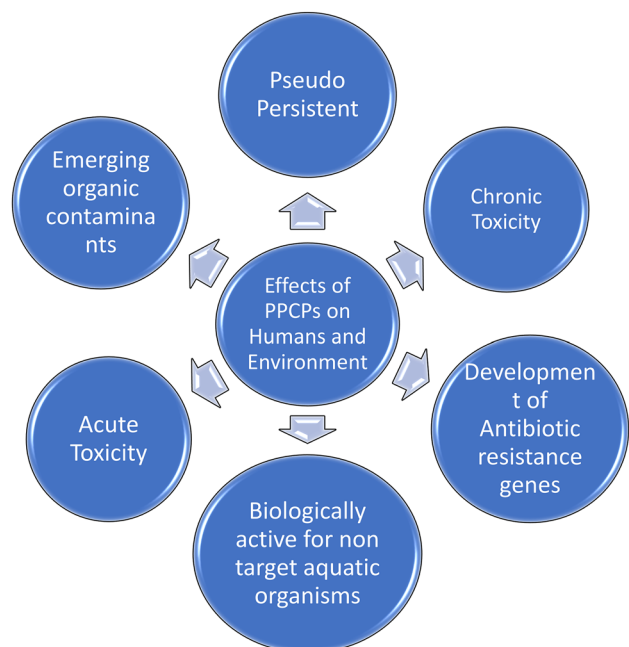
Environmental effects of PPCPs

The presence of PPCPs in the ecosystem is a major concern throughout the globe because of their possible toxicological impacts on all living beings (Ávila & García, 2015). However, the information about the environmental toxicity of PPCPs is still inadequate, principally for prolonged exposure to low concentrations (Al-Baldawi et al., 2021). Together with toxicity, persistence is an additional worrying aspect to be kept in mind when studying the potential effects of PPCPs (Dordio & Carvalho, 2018).

Therefore, even though the accumulation of some PPCPs is not noteworthy, they might be persisting because of their constant release into the ecosystems (Ávila & García, 2015). It is also reported that several PPCPs are biologically active for non-target organisms, such as mosquito fish, goldfish, and the algae *Pseudokirchneriella subcapitata* (Ebele et al., 2017). Several PPCPs have the potential to cause acute toxicity, while others may pose indirect effects on various aquatic species (Button et al., 2019). Pharmaceuticals such as ibuprofen, diclofenac, carbamazepine, and fluoxetine cause acute toxicity in algae, bacteria, *Daphnia*, and other higher trophic level animals such as fishes. However, several other antibiotics have been recognized to possess toxic effects on aquatic organisms at both lower and higher concentrations (Patel et al., 2019). It is reported that chloroquine and hydroxychloroquine both pose significant toxicological hazards in humans through therapeutic uses, unsupervised administration, and overdose. This leads to acute toxicity with a 10–30% mortality rate due to cardiovascular effects and electrolyte imbalances (Della Porta et al., 2020). Liu et al. (2018b) also highlighted the possible adverse effects of erythromycin by calculating the risk quotient (RQ) of the most sensitive algal group in Nanjing, China. They described that the RQ values of erythromycin were found higher which can be responsible for both acute and

chronic toxicity. In addition, the high intake of antimicrobial agents causes a substantial accumulation of these substances facilitating the formation of antibiotic resistance genes (Roca et al., 2015). Nowrotek et al. (2016) explained that both antibiotic-resistant bacteria and genes (ARB and ARGs) are considered emergent ecological pollutants, with possible consequences on humans as well as on the ecosystem. Presently, most of the PPCPs are not listed in the category of the regulated substance; therefore, there are no discharge standards (Ewadh et al., 2017). However, it is proposed by several researchers that some priority PPCPs that pose great threats to living creatures might be screened and regulated (Junaid et al., 2019; Lyu et al., 2019). Various countries have been framing guidelines for the proper management of PPCPs. For that, the US Center for Drug Evaluation and Research (CDER), 1998, issued guidelines for the assessment of associated risk. The US Environmental Protection Agency (USEPA, 2012) also formulated guidelines for pharmaceutical industries to mitigate their effluent release and emission. The European Medicines Agency enacted the primary guiding principles for risk assessment through human medicines in 2006. The agency is also committed to including certain PPCPs within the list of priority contaminants (Bartrons & Peñuelas, 2017), whereas in Switzerland, detailed evidence about the ecotoxicity of

Fig. 2 Potential effects of the PPCPs on well-beings of humans and the environment



pharmaceuticals is mandatory. The possible effects of PPCPs on humans as well as on the environment have been highlighted in Fig. 2.

Fate and removal mechanisms of PPCPs

The fate of PPCPs within the environment is primarily reliant on the characteristics of these contaminants such as chemical structure and acidity constant and the existing environmental conditions. Several methods such as biodegradation, sorption, photodegradation, volatilization, oxidation, and hydrolysis take place during the phytoremediation of PPCPs within CWs (Gorito et al., 2017). Phytoremediation includes the use of macrophytes and associated microbial communities present in the rhizosphere zone for the exclusion or reduction of pollutants (Abdullah et al., 2020). Macrophytes have been extensively utilized for the reduction, metabolization, or sequestering of inorganic and organic pollutants from wastewaters throughout the world (Kotzya et al., 2010). However, their effectiveness in eliminating PPCPs is mainly based on their physicochemical characteristics like lipophilicity and hydrophilicity (Wang et al., 2019). Phytoremediation also provides aesthetic value to the surroundings together with wastewater treatment and several environmental benefits (Zhang et al., 2010). Currently, researchers throughout the world are focusing on green solutions for the treatment of wastewaters using plants and microorganisms (Abd Kadir et al., 2020). Furthermore, macrophytes can also tolerate higher concentrations of PPCPs.

CWs for the removal of PPCPs

CWs are artificially engineered ecosystems that are primarily used for wastewater treatment in specific environments (Kumar & Dutta, 2019). CWs mainly comprise of macrophytes, substrate materials, and microorganisms to utilize natural processes during the treatment (Kumar et al., 2021; Wan et al., 2021; Wu et al., 2015). The selection of substrate material is very crucial as it facilitates the growth of macrophytes, adsorption of PPCPs, and adherence sites for microbial growth (Francini et al., 2018; Yang et al., 2018). Macrophytes within CWs remove PPCPs, either through direct uptake or by creating favorable conditions to eliminate them. PPCPs may also be subject to phytodegradation

via metabolic pathways. Furthermore, macrophytes also support the microbial breakdown of PPCPs by offering the surface for attachment of microbial populations. Consequently, the PPCPs go through a transformation into hydrophilic and/or hydrophobic compounds (Dhir, 2019). However, a known amount of the contaminants is retained in the substrate material. The adsorption of contaminants into the substrate is greatly dependent on the substrate materials and the characteristics of compounds (Dhir, 2019). The use of CWs for the removal of PPCPs has been studied extensively throughout the world even for those that are recalcitrant to the traditional treatment approaches (Chen et al., 2018; Dordio & Carvalho, 2018; Ferreira et al., 2017). They can be utilized for the elimination of a range of micropollutants, such as reactive oxygen species (ROS), and to reduce the anti-androgen (Ant-AR) activities and cytotoxicity as described by Xu et al. (2019b). CWs are considered environmentally friendly as compared to other conventional wastewater treatment technologies together with several advantages, such as lower energy, operation and maintenance requirements, groundwater recharge, recreation uses, aesthetic values, and flood control (Kumar & Dutta, 2019; Liang et al., 2018). However, the macrophyte biomass generated as by-products may be harvested and utilized for the production of compost or biofuels. Several recent articles have measured the exclusion of PPCPs through CWs in different ways. Studies carried out by Ilyas and van Hullebusch (2020a) have evaluated the efficacy of CWs for the removal of pharmaceutical substances, impact of aeration, and by-products formed during the treatment time. Further, they have also studied the role of design and working parameters, elimination mechanism, and the consequences of seasons for the removal of PPCPs (Ilyas & van Hullebusch, 2020b). A comprehensive study on the elimination mechanisms of PPCPs through CWs and associated ecological risks was also performed by Ilyas et al. (2020). Furthermore, the removal of antibiotic resistance genes, COD, nitrogen, and phosphorus were studied by Liu et al. (2019b). However, the cumulative effect of plants and bacteria on the elimination of PPCPs was studied by Nguyen et al. (2019). In another study, the link between the availability of PPCPs and working conditions was also studied by Rabello et al. (2019). The biogeochemical responses distress the elimination of PPCPs mainly via the oxidation–reduction (redox) methods and the processes taking place in the rhizosphere zone (Imfeld

et al., 2009). Several studies advised that the oxidation–reduction potential (ORP) is the key aspect influencing the removal of PPCPs (Huang et al., 2004; Matamoros et al., 2012). Aerobic degradation of ibuprofen, sulfamethoxazole, and salicylic acid is considered the main contributor through microbial degradation, while anaerobic degradation was supposed to be accountable for the biodegradation of caffeine and naproxen (Li et al., 2014). Several studies have evaluated that high ORP may encourage the exclusion of diclofenac, whereas some other findings advised that the elimination rate might be improved under anaerobic settings (Ávila et al., 2010, 2013; Hijosa-Valsero et al., 2011). Song et al. (2009) suggested that the biodegradation of estrogens occurs effectively under aerobic settings for both solid and liquid phases as compared to anoxic or anaerobic conditions. However, Froehner et al. (2011) described optimum elimination of estrogens under both anaerobic and aerobic circumstances.

Key parameters responsible for altering the performance of CWs

The exclusion of contaminants in CWs proceeds via several physicochemical methods and microbial interactions (Ávila & García, 2015). The successful operation of CWs needs the careful measure of numerous parameters such as water depth, wetland configuration, suitable macrophytes and substrate materials, feeding mode, hydraulic loading rate (HLR), and appropriate hydraulic retention time (HRT) (Gorito et al., 2017; Kumar & Dutta, 2019; Wu et al., 2015). The removal efficacy of CWs is also influenced greatly by several prevailing ecological conditions such as pH, dissolved oxygen (DO), and temperature (Dordio & Carvalho, 2018). Nevertheless, very limited research studies are reported that deals with the optimization of design criteria (Kaur et al., 2019; Stefanakis, 2018; Zhang et al., 2018b). It is well explained that the pH of the wastewater is considered a crucial parameter as it facilitates numerous biotic processes (Hijosa-Valsero et al., 2010). The optimum pH values ranged from near neutral to neutral for the better growth of macrophytes as well as for the activity of nitrifying and heterotrophic bacteria within CWs (Dordio & Carvalho, 2013; Ilyas & van Hullebusch, 2019). The high influent pH affects the adsorption capacity as it distresses the dissociation of PPCPs and their successive attachment to the substrate materials by ion exchange (Auvinen et al.,

2017). It is also reported that higher temperature favors the endothermic reaction and accelerates the biodegradation of various PPCPs (Truu et al., 2009). The majority of pharmaceuticals have a long half-life and are hydrolytically stable. Therefore, microbial degradation is the probably dominant pathway for their elimination in CWs (Hijosa-Valsero et al., 2010). The optimum temperature within CWs for microbial metabolism ranges from 15 to 25 °C as explained in previous studies (Koottatep et al., 2017; Ren et al., 2021). The growth of macrophytes is enhanced in summers which ultimately improves the uptake capacity and transportation of PPCPs. The oxygen transport ability of macrophyte's roots is also significantly heightened, which accelerates the decomposition and oxidation of contaminants in CWs (Patel et al., 2019). Ilyas and van Hullebusch (2019) recorded a higher elimination rate for several pharmaceuticals such as diclofenac, naproxen, sulfadiazine, ketoprofen, caffeine, and salicylic acid in summer as compared to in winter. Several other factors also affect the elimination of PPCPs, in which the presence of organic matter is the main one that can influence the adsorption capability (Dordio & Carvalho, 2013; Lee, 2013). Sharif et al. (2014) evaluated that the presence of excess organic matter can reduce the photodegradation of various PPCPs. The number and type of PPCPs in the influent can also disturb the exclusion mechanisms due to interactions (Vo et al., 2018). Several physicochemical parameters such as pH, temperature, and ionic strength can also affect the adsorption and desorption of PPCPs (Ávila et al., 2017).

Rhizosphere community

The interaction of microbial communities with wetland components provides vital information about the biodegradation of contaminants (Zhang et al., 2016). Identification and characterization of microbial populations and the associated metabolic reactions are providing further insights for optimizing the environmental and operational conditions (Hijosa-Valsero et al., 2018). Numerous studies have been done extensively on the elimination of PPCPs through CWs; however, the data on the interaction of macrophytes and microorganisms for the biodegradation of PPCP is still inadequate (Hussain et al., 2018; Nguyen et al., 2019). It is reported that the CWs planted with *Phragmites australis* revealed higher microbial populations

as compared to *Typha angustifolia* (Hijosa-Valsero et al., 2018). Similar results were also observed in a study dealing with domestic wastewater treatment using these macrophytes in respect of enzyme activities (Kumar et al., 2020). Liu et al. (2016) observed that the CWs planted with *Typha sp.* exhibited higher microbial richness and diversity as compared to *Ceratophyllum demersum* and *Lemna minor*. Therefore, understanding the structure and diversity of microbial populations present in a CW may help to improve their performance (Zhang et al., 2016). It is observed that a range of microorganisms such as *Dechloromonas sp.*, *Clostridium sp.*, *Cytophagales*, *Sphingobacteriales*, and *Cytophaga sp.* are responsible for the biodegradation of ibuprofen under anaerobic settings (Li et al., 2016). It is also estimated that the transformation of sulfamethoxazole occurs in the presence of glutathione within the CW as it acts as a marker of oxidative stress in microorganisms and macrophytes (Nowrotek et al., 2016). In addition to these, the continuous exposure of antibacterial contaminants to microbial communities may reduce the efficacy of wetland systems (Button et al., 2019). Nevertheless, various types of bacteria, such as *Bacteroidetes* and β and γ -*Proteobacteria* have a significant role in triclosan degradation within the CW (Liu et al., 2016). Removal efficacy of various PPCPs together with associated mechanisms has been provided in Table 1.

SWOT analysis on CW systems for PPCP removal

The SWOT analysis of CWs provides an excellent option for further improvement in the performance and their sustainable working on a field scale (Nabi Bidhendi et al., 2020). Currently, researchers throughout the globe have significant concerns about SWOT study to prioritize the technologies for wastewater treatment. They summarize and analyze the data for wastewater remediation technologies' social, economic, and environmental prospects (Kalabić et al., 2019). Consequently, the SWOT examination of this review article is mainly focusing on the several attributes linked with CWs for the elimination of PPCPs as exhibited in Fig. 3 (Al-Baldawi et al., 2021). The strengths define positive attributes, and the weaknesses are the aspects missing during the working of CWs. Opportunities are considered exterior positive aspects that signify the reasons to focus on CWs for PPCP removal. However, threats may be explained as

negative characteristics that can influence the efficacy of wetland systems. CWs have emerged as an efficient technology for the elimination of a range of pollutants including PPCPs due to their eco-friendliness and sustainability over other conventional technologies (Abdullah et al., 2020). Furthermore, they are usually easy to handle and provide a cost-effective solution for wastewater reclamation polluted with various PPCPs. CWs can provide several other advantages such as obtained macrophytic biomass may be used as raw material in pulp and paper industries and for second-generation biofuel production (Lin et al., 2020). However, to acquire these benefits from CWs together with wastewater treatment, more research is needed in upcoming studies. The use of CWs has attained notable outcomes in eliminating PPCPs, excluding some recalcitrant compounds. Generally, they need higher HRT as compared to other conventional methods. Moreover, seasonal variations and prevailing environmental conditions may also inhibit the growth of macrophytes (Lee, 2013). To overcome this weakness, intermittent aeration and the addition of efficient rhizobacteria, biosurfactants, and enzymes may provide a good opportunity to accelerate the removal of PPCPs. In terms of opportunities, CWs can also be integrated with wastewater treatment plants as a polishing unit to eliminate remaining PPCPs to meet discharge values. After the polishing in the tertiary unit, the recycled water may be reused for several other utilities such as non-potable household activities, landscape gardening, and industrial processing. However, the intensified wetlands can be applied to enhance the exclusion efficacy and also lessen overall land necessities. The threat associated with CWs during the removal of PPCPs is the probability of the development of more toxic by-products that might be buildup in the food chain. Assessing the effectiveness of CWs for a long period can offer valuable information as described by Tondera et al. (2019). However, a change of the substrate material or a successive filter step is essential after long-term operation to sustain maximum PPCP removal (Tondera et al., 2019; Zhang et al., 2014).

Future concerns and challenges

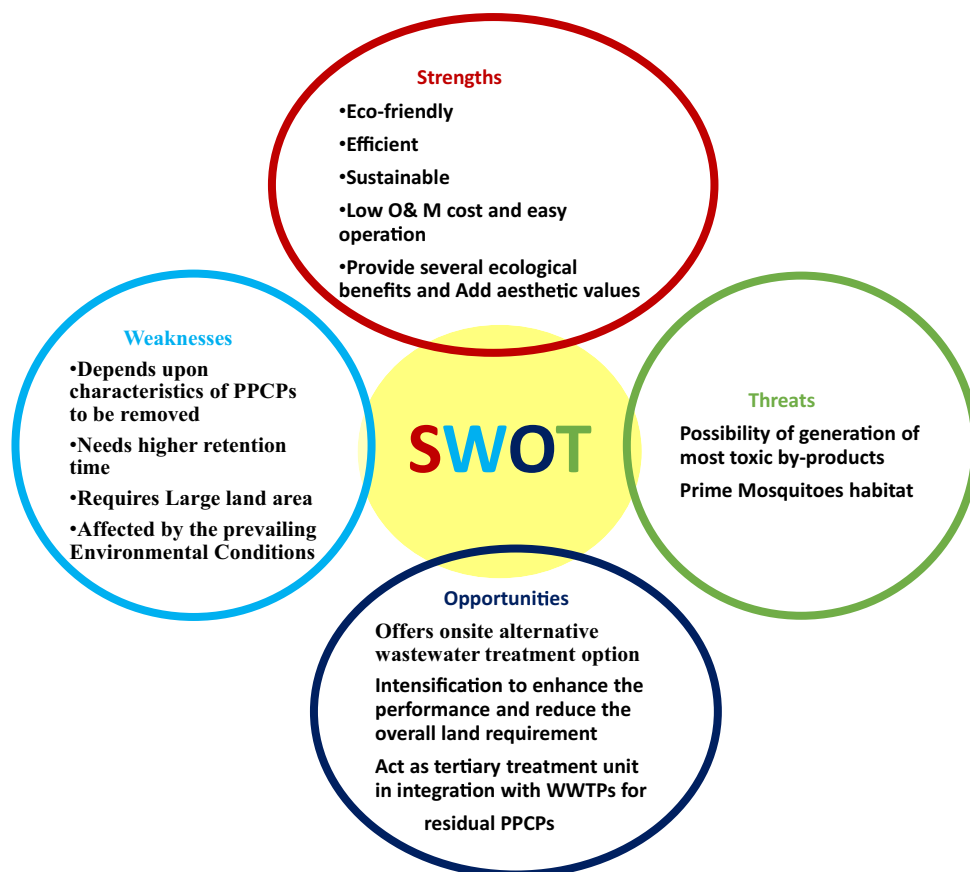
CWs possess great possibilities for the exclusion of PPCPs. However, the exclusion of refractory PPCPs such as ibuprofen, carbamazepine, and diclofenac pose significant challenges (Abdullah et al., 2020). For that, studies on the metabolic pathways of PPCPs

Table 1 Classification of PPCPs, their functions, removal mechanisms, and efficacy as reported in the literature

Classification	PPCPs	Functions	Molecular weight	Log K _{ow}	Removal mechanisms	Removal percentage	Reference
Antibiotics	Sulfamethoxazole	Kill bacteria, treat fever and mild to moderate pain	253.3	0.89	Adsorption	69	Sochacki et al. (2018)
	Ofloxacin		361.4	−0.39		90.75	Yan (2014)
	Roxithromycin		837.0	2.75	Adsorption	85.35	Yan (2014)
	Azithromycin		748.99				Zhang et al. (2014)
	Paracetamol		151.163		Sorption and biodegradation		Chen et al. (2016)
NSAID (non-steroidal anti-inflammatory drugs)	Ibuprofen	Reduce pain and inflammation	206.3	3.97	Biodegradation, Plant uptake	78	He et al. (2018)
					83	Zhang et al. (2015)	
	Diclofenac		296.1	4.50	Biodegradation	74.20	Sochacki et al. (2018)
	Ketoprofen		254.3	2.80		98.20	Zhang et al. (2015); Chen et al. (2016)
	Naproxen	230.3	3.18	Biodegradation	62.07	Hijosa-Valsero et al. (2016)	
					90	He et al. (2018)	
					93	Zhang et al. (2015)	
	Salicylic acid	138.1	2.26	Plant uptake	94.03	Hijosa-Valsero et al. (2016)	
					94.20	Zhang et al. (2015)	
Cosmetics	Methyl dihydro-jasmonate	Make-up and skin-care	226.3	2.98	-	87.07	Hijosa-Valsero et al. (2016)
Fragrances	Galaxolide	Flavoring essence	258.4	5.29	-	68.03	Hijosa-Valsero et al. (2016)
	Tonalide		258.4	5.78	-	72	Hijosa-Valsero et al. (2016)
Endocrine disruptor	Bisphenol A	Affect endocrine system	228.3	3.32	Photodegradation	73.30	Toro-Vélez et al. (2016)
Surfactants	Linear alkylbenzene sulphonate (LAS)	Surface tension and free energy reduction	-		Adsorption	99.60	Thomas et al. (2017)
Beta-blockers	Propranolol	Inhibit the role of the noradrenalin and adrenalin	259.3	3.48	Adsorption, Photodegradation	85	He et al. (2018)
	Metoprolol		267.4	9.70	Biodegradation, Photodegradation	86	He et al. (2018)
Lipid-reducing agents	Clofibrilic acid	Regulates the cholesterol and triglycerides in the blood	214.7	2.57		39	Zhang et al. (2015)
Anticonvulsants	Carbamazepine	Treat mood disorders	236.3	13.90	Photodegradation and plant uptake		He et al. (2018)
Stimulants	Caffeine	Stimulates the central nervous system	194.2	−0.07		95.73	Hijosa-Valsero et al. (2016)
						99	He et al. (2018)
						93.30	Zhang et al. (2015)

Table 1 (continued)

Classification	PPCPs	Functions	Molecular weight	Log K_{ow}	Removal mechanisms	Removal percentage	Reference
Antibacterial and antifungal agents	Triclosan	Reduce or prevent bacterial contamination	289.54	-	Biodegradation, Adsorption	-	Ávila and García (2015)
Repellent	DEET	Protects against mosquitoes, ticks, fleas, chiggers, leeches and many biting insects	191.274	-	Photodegradation	-	Li et al. (2017)
Antidepressant	Sertraline	Treat depression, and also panic attacks sometimes	306.229		Plant uptake		Nuel et al. (2018)

**Fig. 3** SWOT study demonstration of CWs for the treatment of wastewater laden with PPCPs (modified from Al-Baldawi et al., 2021)

and the rhizosphere effect might be significantly important towards understanding their removal mechanisms. Some future research might also be focused on translocation and absorption of lower-concentration hydrophilic and hydrophobic PPCPs. The contribution of different macrophytic tissues to the removal of PPCPs may be assessed. However, more research is needed on the identification of metabolites of PPCPs within macrophytic tissues, which are crucial to explaining the underlying mechanisms. The information on the interface of macrophytes and microorganisms for the degradation of PPCP is also largely inadequate. Together with these, the growth pattern of macrophytes within CWs may also be explored further. Proper harvesting of macrophytes may reduce the concentration of secondary contaminants originating from the transformation of PPCPs. Generally, CWs require more time as compared to other treatment methods. Moreover, seasonal variations and prevailing environmental conditions may also inhibit the growth of macrophytes (Al-Baldawi et al., 2021; Lee, 2013). To overcome this weakness, intermittent aeration and the addition of efficient rhizobacteria, biosurfactants, and enzymes may provide a good opportunity to accelerate the removal of PPCPs.

Conclusion

The continuous exposure of PPCPs can lead to successive individual antagonistic and synergistic effects and also long-lasting toxicity to humans as well as aquatic life. Several PPCPs are considered biologically active for non-target aquatic fauna, such as mosquito fish, goldfish, and *Pseudokirchneriella subcapitata*. However, the fate of PPCPs in the ecosystems primarily relied on their chemical structure, acidity constant, octanol coefficient, etc., together with existing environmental settings. Several physicochemical and biotechnological methods such as ozonation, photolysis, and membrane separation are identified for the effective removal of PPCPs. Nevertheless, the high operation and maintenance costs and logistical aspects have limited their use on a large scale. CWs are considered sustainable and more suitable for the treatment of different types of PPCPs together with antibiotic resistance genes. Under optimized conditions, more than 50% of the total PPCPs can be eliminated through CW systems. Macrophytes remove PPCPs through direct

uptake or by creating favorable environments for microbial transformation by offering attachment sites. Furthermore, PPCPs may also be subject to phytodegradation via metabolic actions. However, the number and type of PPCPs in the influent can disturb the exclusion mechanisms due to interactions among them. CWs also offer several other advantages over conventional methods as resulting macrophytic biomass may be used as raw material in pulp and paper industries and for second-generation biofuel production. They can also be integrated with wastewater treatment plants as a polishing unit to eliminate remaining PPCPs to meet discharge values. The recycled water may also be reused for several other utilities such as non-potable household activities, agricultural irrigation, landscape gardening, and industrial processing.

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Declarations

Competing interests The authors declare no competing interests.

References

- Abd Kadir, A., Abdullah, S. R. S., Othman, B. A., Hasan, H. A., Othman, A. R., Imron, M. F., & Kurniawan, S. B. (2020). The dual function of *Lemna minor* and *Azolla pinnata* as phytoremediators for palm oil mill effluent and feedstock. *Chemosphere*, 259, 127468.
- Abdullah, S. R. S., Al-Baldawi, I. A., Almansoori, A. F., Purwanti, I. F., Al-Sbani, N. H., & Sharuddin, S. S. N. (2020). Plant-assisted remediation of hydrocarbons in water and soil: Application, mechanisms, challenges and opportunities. *Chemosphere*, 247, 125932.
- Al-Baldawi, I. A., Mohammed, A. A., Mutar, Z. H., Abdullah, S. R. S., Jasim, S. S., & Almansoori, A. F. (2021). Application of phytotechnology in alleviating pharmaceuticals and personal care products (PPCPs) in wastewater: Source, impacts, treatment, mechanisms, fate, and SWOT analysis. *Journal of Cleaner Production*, 319, 128584.
- Alhaji, N. B., & Isola, T. O. (2018). Antimicrobial usage by pastoralists in food animals in North-central Nigeria: The associated socio-cultural drivers for antimicrobials misuse and public health implications. *One Health*, 6, 41–47.
- Archer, E., Petrie, B., Kasprzyk-Hordern, B., & Wolfaardt, G. M. (2017). The fate of pharmaceuticals and personal care products (PPCPs), endocrine-disrupting contaminants (EDCs), metabolites and illicit drugs in a WWTW and environmental waters. *Chemosphere*, 174, 437–446.

- Auvinen, H., Gebhardt, W., Linnemann, V., Du Laing, G., & Rousseau, D. P. (2017). Laboratory-and full-scale studies on the removal of pharmaceuticals in an aerated constructed wetland: Effects of aeration and hydraulic retention time on the removal efficiency and assessment of the aquatic risk. *Water Science and Technology*, 76(6), 1457–1465.
- Ávila, C., & García, J. (2015). Pharmaceuticals and personal care products (PPCPs) in the environment and their removal from wastewater through constructed wetlands. In *Comprehensive analytical chemistry* (Vol. 67, pp. 195–244). Elsevier.
- Ávila, C., Pedescoll, A., Matamoros, V., Bayona, J. M., & García, J. (2010). Capacity of a horizontal subsurface flow constructed wetland system for the removal of emerging pollutants: An injection experiment. *Chemosphere*, 81, 1137–1142.
- Ávila, C., Pelissari, C., Sezerino, P. H., Sgroi, M., Roccaro, P., & García, J. (2017). Enhancement of total nitrogen removal through effluent recirculation and fate of PPCPs in a hybrid constructed wetland system treating urban wastewater. *Science of the Total Environment*, 584, 414–425.
- Ávila, C., Reyes, C., Bayona, J. M., & García, J. (2013). Emerging organic contaminant removal depending on primary treatment and operational strategy in horizontal subsurface flow constructed wetlands: Influence of redox. *Water Research*, 47(1), 315–325.
- Bartrons, M., & Peñuelas, J. (2017). Pharmaceuticals and personal care products in plants. *Trends in Plant Science*, 22(3), 194–203.
- Bi, R., Zhou, C., Jia, Y., Wang, S., Li, P., Reichwaldt, E. S., & Liu, W. (2019). Giving waterbodies the treatment, they need: A critical review of the application of constructed floating wetlands. *Journal of Environmental Management*, 238, 484–498.
- Button, M., Cosway, K., Sui, J., & Weber, K. (2019). Impacts and fate of triclosan and sulfamethoxazole in intensified re-circulating vertical flow constructed wetlands. *Science of the Total Environment*, 649, 1017–1028.
- Chaves, M. D. J. S., Barbosa, S. C., de Melo Malinowski, M., Volpato, D., Castro, Í. B., dos Santos Franco, T. C. R., & Primel, E. G. (2020). Pharmaceuticals and personal care products in a Brazilian wetland of international importance: Occurrence and environmental risk assessment. *Science of the Total Environment*, 734, 139374.
- Chen, X., Hu, Z., Zhang, Y., Zhuang, L., Zhang, J., Li, J., & Hu, H. (2018). Removal processes of carbamazepine in constructed wetlands treating secondary effluent: A review. *Water*, 10(10), 1351.
- Chen, Y., Vymazal, J., Březinová, T., Koželuh, M., Kule, L., Huang, J., & Chen, Z. (2016). Occurrence, removal and environmental risk assessment of pharmaceuticals and personal care products in rural wastewater treatment wetlands. *Science of the Total Environment*, 566, 1660–1669.
- Chopra, S., & Kumar, D. (2018). Pharmaceuticals and personal care products (PPCPs) as emerging environmental pollutants: Toxicity and risk assessment. In *Advances in animal biotechnology and its applications* (pp. 337–353). Springer, Singapore.
- Della Porta, A., Bornstein, K., Coye, A., Montrieff, T., Long, B., & Parris, M. A. (2020). Acute chloroquine and hydroxychloroquine toxicity: A review for emergency clinicians. *The American Journal of Emergency Medicine*, 38(10), 2209–2217.
- Dey, S., Bano, F., & Malik, A. (2019). Pharmaceuticals and personal care product (PPCP) contamination—A global discharge inventory. In *Pharmaceuticals and personal care products: waste management and treatment technology* (pp. 1–26). Butterworth-Heinemann.
- Dhir, B. (2019). Removal of pharmaceuticals and personal care products by aquatic plants. In *Pharmaceuticals and personal care products: Waste management and treatment technology* (pp. 321–340). Butterworth-Heinemann.
- Dordio, A., & Carvalho, A. J. P. (2013). Constructed wetlands with light expanded clay aggregates for agricultural wastewater treatment. *Science of the Total Environment*, 463, 454–461.
- Dordio, A., & Carvalho, A. J. P. (2018). Removal processes of pharmaceuticals in constructed wetlands. *Constructed Wetlands for Industrial Wastewater Treatment*, 343–403.
- Ebele, A. J., Abdallah, M. A. E., & Harrad, S. (2017). Pharmaceuticals and personal care products (PPCPs) in the freshwater aquatic environment. *Emerging Contaminants*, 3(1), 1–16.
- Eggen, T., & Vogelsang, C. (2015). Occurrence and fate of pharmaceuticals and personal care products in wastewater. In *Comprehensive Analytical Chemistry* (Vol. 67, pp. 245–294). Elsevier.
- Ewadh, H. M., Abdullah, S. R., Anwar, N., & Hasan, H. A. (2017). Pharmaceuticals and personal care products: Sources, toxicity in the environment, regulations and removal technologies. *Journal of Chemical and Pharmaceutical Sciences*, 10, 1180–1187.
- Fadhil, N. M., & Al Baldawi, I. A. W. (2020). Biodegradation of total petroleum hydrocarbon from Al-Daura refinery wastewater by rhizobacteria. *Journal of Engineering*, 26(1), 14–23.
- Ferreira, A. R., Ribeiro, A., & Couto, N. (2017). Remediation of pharmaceutical and personal care products (PPCPs) in constructed wetlands: Applicability and new perspectives. In *Phytoremediation* (pp. 277–292). Springer, Cham.
- Francini, A., Mariotti, L., Di Gregorio, S., Sebastiani, L., & Andreucci, A. (2018). Removal of micropollutants from urban wastewater by constructed wetlands with *Phragmites australis* and *Salix matsudana*. *Environmental Science and Pollution Research*, 25(36), 36474–36484.
- Froehner, S., Piccioni, W., Machado, K. S., & Aisse, M. M. (2011). Removal capacity of caffeine, hormones, and bisphenol by aerobic and anaerobic sewage treatment. *Water, Air, & Soil Pollution*, 216(1), 463–471.
- Gorito, A. M., Ribeiro, A. R., Almeida, C. M. R., & Silva, A. M. (2017). A review on the application of constructed wetlands for the removal of priority substances and contaminants of emerging concern listed in recently launched EU legislation. *Environmental Pollution*, 227, 428–443.
- He, Y., Sutton, N. B., Lei, Y., Rijnaarts, H. H., & Langenhoff, A. A. (2018). Fate and distribution of pharmaceutically

- active compounds in mesocosm constructed wetlands. *Journal of Hazardous Materials*, 357, 198–206.
- Hijosa-Valsero, M., Fink, G., Schlüsener, M. P., Sidrach-Cardona, R., Martín-Villacorta, J., Ternes, T., & Bécares, E. (2011). Removal of antibiotics from urban wastewater by constructed wetland optimization. *Chemosphere*, 83(5), 713–719.
- Hijosa-Valsero, M., Matamoros, V., Sidrach-Cardona, R., Martín-Villacorta, J., Bécares, E., & Bayona, J. M. (2010). Comprehensive assessment of the design configuration of constructed wetlands for the removal of pharmaceuticals and personal care products from urban wastewaters. *Water Research*, 44(12), 3669–3678.
- Hijosa-Valsero, M., Reyes-Contreras, C., Domínguez, C., Bécares, E., & Bayona, J. M. (2016). Behaviour of pharmaceuticals and personal care products in constructed wetland compartments: Influent, effluent, pore water, substrate and plant roots. *Chemosphere*, 145, 508–517.
- Hijosa-Valsero, M., Sidrach-Cardona, R., Pedescoll, A., Sánchez, O., & Bécares, E. (2018). Role of bacterial diversity on PPCPs removal in constructed wetlands. *Constructed Wetlands for Industrial Wastewater Treatment*, 405–426.
- Hoyett, Z. (2018). Pharmaceuticals and personal care products: Risks, challenges and solutions. *Risk Assessment*.
- Hu, X., Xie, H., Zhuang, L., Zhang, J., Hu, Z., Liang, S., & Feng, K. (2021). A review on the role of plant in pharmaceuticals and personal care products (PPCPs) removal in constructed wetlands. *Science of the Total Environment*, 780, 146637.
- Huang, Y., LaTorre, A., Barceló, D., García, J., Aguirre, P., Mujeriego, R., & Bayona, J. M. (2004). Factors affecting linear alkylbenzene sulfonates removal in subsurface flow constructed wetlands. *Environmental Science & Technology*, 38(9), 2657–2663.
- Hussain, I., Aleti, G., Naidu, R., Puschenreiter, M., Mahmood, Q., Rahman, M. M., & Reichenauer, T. G. (2018). Microbe and plant assisted-remediation of organic xenobiotics and its enhancement by genetically modified organisms and recombinant technology: A review. *Science of the Total Environment*, 628, 1582–1599.
- Ilyas, H., & van Hullebusch, E. D. (2019). Role of design and operational factors in the removal of pharmaceuticals by constructed wetlands. *Water*, 11(11), 2356.
- Ilyas, H., & van Hullebusch, E. D. (2020a). Performance comparison of different constructed wetlands designs for the removal of personal care products. *International Journal of Environmental Research and Public Health*, 17(9), 3091.
- Ilyas, H., & van Hullebusch, E. D. (2020b). The influence of design and operational factors on the removal of personal care products by constructed wetlands. *Water*, 12(5), 1367.
- Ilyas, H., Masih, I., & van Hullebusch, E. D. (2020). Pharmaceuticals' removal by constructed wetlands: A critical evaluation and meta-analysis on performance, risk reduction, and role of physicochemical properties on removal mechanisms. *Journal of Water and Health*, 18(3), 253–291.
- Imfeld, G., Braeckvelt, M., Kusch, P., & Richnow, H. H. (2009). Monitoring and assessing processes of organic chemicals removal in constructed wetlands. *Chemosphere*, 74(3), 349–362.
- Jensen, O., & Wu, H. (2018). Urban water security indicators: Development and pilot. *Environmental Science & Policy*, 83, 33–45.
- Junaid, M., Wang, Y., Hamid, N., Deng, S., Li, W. G., & Pei, D. S. (2019). Prioritizing selected PPCPs on the basis of environmental and toxicogenetic concerns: A toxicity estimation to confirmation approach. *Journal of Hazardous Materials*, 380, 120828.
- Kalabić, D., Džarić, G., Džarić, N., & Ikanović, J. (2019). Production of agri-energy crop *Miscanthus giganteus* on land degraded by power industry: SWOT analysis. *Polish Journal of Environmental Studies*, 28(5), 3243–3251.
- Kaur, H., Hippargi, G., Pophali, G. R., & Bansiwala, A. K. (2019). Treatment methods for removal of pharmaceuticals and personal care products from domestic wastewater. In *Pharmaceuticals and Personal Care Products: Waste Management and Treatment Technology* (pp. 129–150). Butterworth-Heinemann.
- Koottatep, T., Vo, H. N. P., Chapagain, S. K., Panuvatvanich, A., Polprasert, C., & Ahn, K. (2017). Performance evaluation of selected aquatic plants and iron-rich media for removal of PPCPs from wastewater in constructed wetlands. *Desalination and Water Treatment*, 91, 281–286.
- K'oreje, K. O., Kandie, F. J., Vergeynst, L., Abira, M. A., Van Langenhove, H., Okoth, M., & Demeestere, K. (2018). Occurrence, fate and removal of pharmaceuticals, personal care products and pesticides in wastewater stabilization ponds and receiving rivers in the Nzoia Basin, Kenya. *Science of the Total Environment*, 637, 336–348.
- Kotzya, J., Soudek, P., Kafka, Z., & Vaněk, T. (2010). Phytoremediation of pharmaceuticals- a preliminary study. *International Journal of Phytoremediation*, 12(3), 306–316.
- Kumar, S., & Dutta, V. (2019). Constructed wetland microcosms as sustainable technology for domestic wastewater treatment: An overview. *Environmental Science and Pollution Research*, 26(12), 11662–11673.
- Kumar, S., Nand, S., Dubey, D., Pratap, B., & Dutta, V. (2020). Variation in extracellular enzyme activities and their influence on the performance of surface-flow constructed wetland microcosms (CWMs). *Chemosphere*, 251, 126377.
- Kumar, S., Nand, S., Pratap, B., Dubey, D., & Dutta, V. (2021). Removal kinetics and treatment efficiency of heavy metals and other wastewater contaminants in a constructed wetland microcosm: Does mixed macrophytic combinations perform better? *Journal of Cleaner Production*, 327, 129468.
- Lee, J. H. (2013). An overview of phytoremediation as a potentially promising technology for environmental pollution control. *Biotechnology and Bioprocess Engineering*, 18(3), 431–439.
- Li, J., Zhou, Q., & Campos, L. C. (2017). Removal of selected emerging PPCP compounds using greater duckweed (*Spirodela polyrrhiza*) based lab-scale free water constructed wetland. *Water Research*, 126, 252–261.
- Li, W. L., Zhang, Z. F., Ma, W. L., Liu, L. Y., Song, W. W., & Li, Y. F. (2018). An evaluation on the intra-day

- dynamics, seasonal variations and removal of selected pharmaceuticals and personal care products from urban wastewater treatment plants. *Science of the Total Environment*, 640, 1139–1147.
- Li, W., Nanaboina, V., Chen, F., & Korshin, G. V. (2016). Removal of polycyclic synthetic musks and antineoplastic drugs in ozonated wastewater: Quantitation based on the data of differential spectroscopy. *Journal of Hazardous Materials*, 304, 242–250.
- Li, Y., Zhang, S., Zhang, W., Xiong, W., Ye, Q., Hou, X., & Wang, P. (2019). Life cycle assessment of advanced wastewater treatment processes: Involving 126 pharmaceuticals and personal care products in life cycle inventory. *Journal of Environmental Management*, 238, 442–450.
- Li, Y., Zhu, G., Ng, W. J., & Tan, S. K. (2014). A review on removing pharmaceutical contaminants from wastewater by constructed wetlands: Design, performance and mechanism. *Science of the Total Environment*, 468, 908–932.
- Liang, Y., Zhu, H., Bañuelos, G., Shutes, B., Yan, B., & Cheng, X. (2018). Removal of sulfamethoxazole from salt-laden wastewater in constructed wetlands affected by plant species, salinity levels and co-existing contaminants. *Chemical Engineering Journal*, 341, 462–470.
- Lin, Y., Zhao, Y., Ruan, X., Barzee, T. J., Zhang, Z., Kong, H., & Zhang, X. (2020). The potential of constructed wetland plants for bioethanol production. *BioEnergy Research*, 13(1), 43–49.
- Liu, J., Dan, X., Lu, G., Shen, J., Wu, D., & Yan, Z. (2018a). Investigation of pharmaceutically active compounds in an urban receiving water: Occurrence, fate and environmental risk assessment. *Ecotoxicology and Environmental Safety*, 154, 214–220.
- Liu, J., Wang, J., Zhao, C., Hay, A. G., Xie, H., & Zhan, J. (2016). Triclosan removal in wetlands constructed with different aquatic plants. *Applied Microbiology and Biotechnology*, 100(3), 1459–1467.
- Liu, M., Yin, H., & Wu, Q. (2019a). Occurrence and health risk assessment of pharmaceutical and personal care products (PPCPs) in tap water of Shanghai. *Ecotoxicology and Environmental Safety*, 183, 109497.
- Liu, X., Guo, X., Liu, Y., Lu, S., Xi, B., Zhang, J., & Bi, B. (2019b). A review on removing antibiotics and antibiotic resistance genes from wastewater by constructed wetlands: Performance and microbial response. *Environmental Pollution*, 254, 112996.
- Liu, Y. L., Wang, X. M., Yang, H. W., & Xie, Y. F. (2018b). Adsorption of pharmaceuticals onto isolated polyamide active layer of NF/RO membranes. *Chemosphere*, 200, 36–47.
- Lyu, S., Chen, W., Qian, J., Wen, X., & Xu, J. (2019). Prioritizing environmental risks of pharmaceuticals and personal care products in reclaimed water on urban green space in Beijing. *Science of the Total Environment*, 697, 133850.
- Matamoros, V., Nguyen, L. X., Arias, C. A., Salvadó, V., & Brix, H. (2012). Evaluation of aquatic plants for removing polar microcontaminants: A microcosm experiment. *Chemosphere*, 88(10), 1257–1264.
- Meng, H., Zhang, X., Zhou, Z., Luo, L., Lan, W., Lin, J. G., Li, X. Y., & Gu, J. D. (2021). Simultaneous occurrence and analysis of both anammox and n-damo bacteria in five full-scale wastewater treatment plants. *International Biodeterioration & Biodegradation*, 156, 105112.
- Mompelat, S., Le Bot, B., & Thomas, O. (2009). Occurrence and fate of pharmaceutical products and by-products, from resource to drinking water. *Environment International*, 35(5), 803–814.
- Nabi Bidhendi, G., Daryabeigi Zand, A., Vaezi Heir, A., & Nabi Bidhendi, A. (2020). Prioritizing of strategies for the ecological design of urban waste transfer stations using SWOT analysis. *Journal of Environmental Science Studies*, 5(2), 2665–2672.
- Ngo, T. H., Van, D. A., Tran, H. L., Nakada, N., Tanaka, H., & Huynh, T. H. (2021). Occurrence of pharmaceutical and personal care products in Cau River, Vietnam. *Environmental Science and Pollution Research*, 28, 12082–12091.
- Nguyen, P. M., Afzal, M., Ullah, I., Shahid, N., Baqar, M., & Arslan, M. (2019). Removal of pharmaceuticals and personal care products using constructed wetlands: Effective plant-bacteria synergism may enhance degradation efficiency. *Environmental Science and Pollution Research*, 26(21), 21109–21126.
- Noguera-Oviedo, K., & Aga, D. S. (2016). Lessons learned from more than two decades of research on emerging contaminants in the environment. *Journal of Hazardous Materials*, 316, 242–251.
- Nowrotek, M., Sochacki, A., Felis, E., & Miksch, K. (2016). Removal of diclofenac and sulfamethoxazole from synthetic municipal wastewater in microcosm downflow constructed wetlands: Start-up results. *International Journal of Phytoremediation*, 18(2), 157–163.
- Nuel, M., Laurent, J., Bois, P., Heintz, D., & Wanko, A. (2018). Seasonal and aging effect on the behavior of 86 drugs in a full-scale surface treatment wetland: Removal efficiencies and distribution in plants and sediments. *Science of the Total Environment*, 615, 1099–1109.
- Ort, C., Van Nuijs, A. L., Berset, J. D., Bijlsma, L., Castiglioni, S., Covaci, A., & Thomas, K. V. (2014). Spatial differences and temporal changes in illicit drug use in Europe quantified by wastewater analysis. *Addiction*, 109(8), 1338–1352.
- Patel, M., Kumar, R., Kishor, K., Mlsna, T., Pittman, C. U., Jr., & Mohan, D. (2019). Pharmaceuticals of emerging concern in aquatic systems: Chemistry, occurrence, effects, and removal methods. *Chemical Reviews*, 119(6), 3510–3673.
- Rabello, V. M., Teixeira, L. C. R. S., Gonçalves, A. P. V., & de Sá Salomão, A. L. (2019). The efficiency of constructed wetlands and algae tanks for the removal of pharmaceuticals and personal care products (PPCPs): A systematic review. *Water, Air, & Soil Pollution*, 230(10), 1–12.
- Rajapaksha, A. U., Premarathna, K. S. D., Gunarathne, V., Ahmed, A., & Vithanage, M. (2019). Sorptive removal of pharmaceutical and personal care products from water and wastewater. In *Pharmaceuticals and Personal Care Products: Waste Management and Treatment Technology* (pp. 213–238). Butterworth-Heinemann.
- Ramírez-Morales, D., Masis-Mora, M., Montiel-Mora, J. R., Cambronero-Heinrichs, J. C., Briceño-Guevara, S., Rojas-Sánchez, C. E., & Rodríguez-Rodríguez, C. E. (2020). Occurrence of pharmaceuticals, hazard

- assessment and ecotoxicological evaluation of wastewater treatment plants in Costa Rica. *Science of the Total Environment*, 746, 141200.
- Ren, X., Zhang, M., Wang, H., Dai, X., & Chen, H. (2021). Removal of personal care products in greywater using membrane bioreactor and constructed wetland methods. *Science of the Total Environment*, 797, 148773.
- Roca, I., Akova, M., Baquero, F., Carlet, J., Cavaleri, M., Coenen, S., & Vila, J. (2015). The global threat of antimicrobial resistance: Science for intervention. *New Microbes and New Infections*, 6, 22–29.
- Rodriguez-Narvaez, O. M., Peralta-Hernandez, J. M., Goonetilleke, A., & Bandala, E. R. (2017). Treatment technologies for emerging contaminants in water: A review. *Chemical Engineering Journal*, 323, 361–380.
- Rout, P. R., Zhang, T. C., Bhunia, P., & Surampalli, R. Y. (2021). Treatment technologies for emerging contaminants in wastewater treatment plants: A review. *Science of the Total Environment*, 753, 141990.
- Sgroi, M., Pelissari, C., Roccaro, P., Sezerino, P. H., García, J., Vagliasindi, F. G., & Ávila, C. (2018). Removal of organic carbon, nitrogen, emerging contaminants and fluorescing organic matter in different constructed wetland configurations. *Chemical Engineering Journal*, 332, 619–627.
- Sharif, F., Westerhoff, P., & Herckes, P. (2014). Impact of hydraulic and carbon loading rates of constructed wetlands on contaminants of emerging concern (CECs) removal. *Environmental Pollution*, 185, 107–115.
- Sharma, B. M., Bečanová, J., Scheringer, M., Sharma, A., Bharat, G. K., Whitehead, P. G., & Nizzetto, L. (2019). Health and ecological risk assessment of emerging contaminants (pharmaceuticals, personal care products, and artificial sweeteners) in surface and groundwater (drinking water) in the Ganges River Basin, India. *Science of the Total Environment*, 646, 1459–1467.
- Shetty, A., & Gupta, G. (2018). Design methodologies for eco-friendly pharmaceutical waste management-A review. In *International Conference on Sustainable Waste Management through Design* (pp. 586–595). Springer, Cham.
- Sochacki, A., Felis, E., Bajkacz, S., Nowrotek, M., & Miksch, K. (2018). Removal and transformations of diclofenac and sulfamethoxazole in a two-stage constructed wetland system. *Ecological Engineering*, 122, 159–168.
- Song, H. L., Nakano, K., Taniguchi, T., Nomura, M., & Nishimura, O. (2009). Estrogen removal from treated municipal effluent in small-scale constructed wetland with different depth. *Bioresource Technology*, 100(12), 2945–2951.
- Stefanakis, A. I. (2018). Constructed Wetlands case studies for the treatment of water polluted with fuel and oil hydrocarbons. In *Phytoremediation* (pp. 151–167). Springer, Cham.
- Sutar, R. S., Motghare, V. M., Kollur, S. C., Parikh, Y., & Asolekar, S. R. (2019). Significance of addressing persistence of pathogens and micropollutants to enhance reuse of treated sewage using constructed wetlands. In *Advances in Waste Management* (pp. 355–367). Springer, Singapore.
- Tarpani, R. R. Z., & Azapagic, A. (2018). Life cycle environmental impacts of advanced wastewater treatment techniques for removal of pharmaceuticals and personal care products (PPCPs). *Journal of Environmental Management*, 215, 258–272.
- Thomas, K. V., Bijlsma, L., Castiglioni, S., Covaci, A., Emke, E., Grabic, R., & de Voogt, P. (2012). Comparing illicit drug use in 19 European cities through sewage analysis. *Science of the Total Environment*, 432, 432–439.
- Thomas, R., Gough, R., & Freeman, C. (2017). Linear alkylbenzene sulfonate (LAS) removal in constructed wetlands: The role of plants in the treatment of a typical pharmaceutical and personal care product. *Ecological Engineering*, 106, 415–422.
- Tondera, K., Ruppelt, J. P., Pinnekamp, J., Kistemann, T., & Schreiber, C. (2019). Reduction of micropollutants and bacteria in a constructed wetland for combined sewer overflow treatment after 7 and 10 years of operation. *Science of the Total Environment*, 651, 917–927.
- Toro-Vélez, A. F., Madera-Parra, C. A., Peña-Varón, M. R., Lee, W. Y., Bezares-Cruz, J. C., Walker, W. S., & Lens, P. N. L. (2016). BPA and NP removal from municipal wastewater by tropical horizontal subsurface constructed wetlands. *Science of the Total Environment*, 542, 93–101.
- Tripathi, S., Singh, V. K., Srivastava, P., Singh, R., Devi, R. S., Kumar, A., & Bhadouria, R. (2020). Phytoremediation of organic pollutants: Current status and future directions. In *Abatement of environmental pollutants* (pp. 81–105). Elsevier.
- Truu, M., Juhanson, J., & Truu, J. (2009). Microbial biomass, activity and community composition in constructed wetlands. *Science of the Total Environment*, 407(13), 3958–3971.
- U.S. Environmental Protection Agency. (2012). Pharmaceuticals and personal care products (PPCP).
- Vo, H. N. P., Bui, X. T., Nguyen, T. M. H., Koottatep, T., & Bandyopadhyay, A. (2018). Insights of the removal mechanisms of pharmaceutical and personal care products in constructed wetlands. *Current Pollution Reports*, 4(2), 93–103.
- Wan, W., Gadd, G. M., Yang, Y. Y., Yuan, W., Gu, J., Ye, L., & Liu, W. (2021). Environmental adaptation is stronger for abundant rather than for rare microorganisms in wetland soils from the Qinghai-Tibet Plateau. *Molecular Ecology*, 30, 2390–2403.
- Wang, Y., Yin, T., Kelly, B. C., & Gin, K. Y. H. (2019). Bioaccumulation behavior of pharmaceuticals and personal care products in a constructed wetland. *Chemosphere*, 222, 275–285.
- Wu, H., Zhang, J., Ngo, H. H., Guo, W., Hu, Z., Liang, S., & Liu, H. (2015). A review on the sustainability of constructed wetlands for wastewater treatment: Design and operation. *Bioresource Technology*, 175, 594–601.
- Xu, J., Xu, Y., Wang, H., Guo, C., Qiu, H., He, Y., & Meng, W. (2015). Occurrence of antibiotics and antibiotic resistance genes in a sewage treatment plant and its effluent-receiving river. *Chemosphere*, 119, 1379–1385.
- Xu, M., Huang, H., Li, N., Li, F., Wang, D., & Luo, Q. (2019a). Occurrence and ecological risk of pharmaceuticals and personal care products (PPCPs) and pesticides in typical surface watersheds, China. *Ecotoxicology and Environmental Safety*, 175, 289–298.

- Xu, S., Zhou, S., Xing, L., Shi, P., Shi, W., Zhou, Q., & Li, A. (2019b). Fate of organic micropollutants and their biological effects in a drinking water source treated by a field-scale constructed wetland. *Science of the Total Environment*, 682, 756–764.
- Yan, Q. (2014). Transportation and distribution of typical pharmaceutically active compounds in the water system of city and their removal in constructed wetland system (CWs). *Doctor Chongqing University*.
- Yang, Y., Ok, Y. S., Kim, K. H., Kwon, E. E., & Tsang, Y. F. (2017). Occurrences and removal of pharmaceuticals and personal care products (PPCPs) in drinking water and water/sewage treatment plants: A review. *Science of the Total Environment*, 596, 303–320.
- Yang, Y., Zhao, Y., Liu, R., & Morgan, D. (2018). Global development of various emerged substrates utilized in constructed wetlands. *Bioresource Technology*, 261, 441–452.
- Zhang, B. Y., Zheng, J. S., & Sharp, R. G. (2010). Phytoremediation in engineered wetlands: Mechanisms and applications. *Procedia Environmental Sciences*, 2, 1315–1325.
- Zhang, D., Gersberg, R. M., Ng, W. J., & Tan, S. K. (2014). Removal of pharmaceuticals and personal care products in aquatic plant-based systems: A review. *Environmental Pollution*, 184, 620–639.
- Zhang, D., Luo, J., Lee, Z. M. P., Maspolim, Y., Gersberg, R. M., Liu, Y., & Ng, W. J. (2016). Characterization of bacterial communities in wetland mesocosms receiving pharmaceutical-enriched wastewater. *Ecological Engineering*, 90, 215–224.
- Zhang, D., Ni, W., Gersberg, R. M., Ng, W. J., & Tan, S. K. (2015). Performance characterization of pharmaceutical removal by horizontal subsurface flow constructed wetlands using multivariate analysis. *Clean-Soil, Air, Water*, 43(8), 1181–1189.
- Zhang, S., Song, H. L., Yang, X. L., Li, H., & Wang, Y. W. (2018a). A system composed of a biofilm electrode reactor and a microbial fuel cell-constructed wetland exhibited efficient sulfamethoxazole removal but induced sul genes. *Bioresource Technology*, 256, 224–231.
- Zhang, X., Jing, R., Feng, X., Dai, Y., Tao, R., Vymazal, J., & Yang, Y. (2018b). Removal of acidic pharmaceuticals by small-scale constructed wetlands using different design configurations. *Science of the Total Environment*, 639, 640–647.

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